

Climate 5000 Series - Gen 4

36K BTU Single Zone Regular Light Commercial System

Ducted IDU

Engineering
 Submittal
 Sheet



BOSCH

Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

36K Btu Single Zone Light Commercial Regular System - Ducted IDU

Indoor part number		8-733-962-713
Indoor model number		BMS500-AAU036-1AHDXD
Outdoor part number		8-733-962-692
Outdoor model number		BMS500-AAS036-1CSXLD
Efficiency Data		AHRI 210/240 2023
SEER2		18.7
EER2		10.1
HSPF2 Region IV		10.2
HSPF2 Region V		7.8
COP(47F)		3.70
COP (5F)		2.10
Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	7900~42500
	Rated Capacity	36000
	Current	15.3
	Input Power	3530
Heating	Capacity Range	8100~43500
	Rated Capacity (47F)	40000
	Rated Capacity (5F)	25600
	Current	14.2
	Input Power	3170
Electrical Data		
Power supply (V / Ø / Hz)		208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)		4.0 / 33.0
Max. fuse indoor/outdoor (A)		15.0 / 35.0
Compressor Data		
Type:		Rotary
Refrigerant oil /oil charge (ml)		VG74 / 1000
Input (W)		2765
Rated current (RLA/A)		5.4
Piping		
Refrigerant precharge (oz/kg)		91.71 / 2.60
Length for precharge (ft/m)		24.6 / 7.5
Liquid side (in/mm)		3/8" / Ø9.52
Gas side (in/mm)		3/4" / Ø19.0
Additional Charge of Refrigerant (oz/ft)/(g/m)		0.32 / 30
Max. refrigerant pipe length (ft/m)		246.0 / 75
Max. difference in height (ft/m)		98.4 / 30

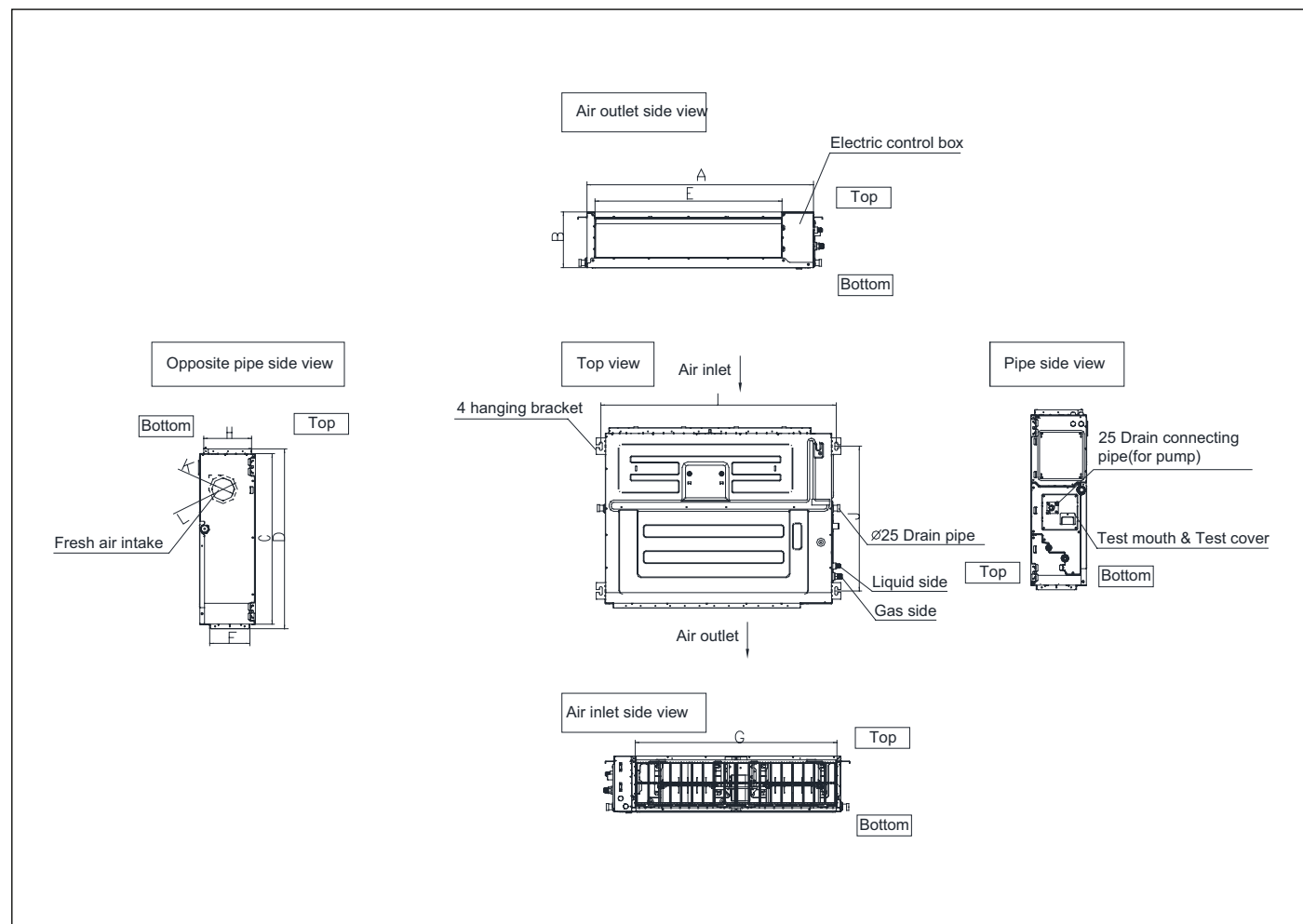
Operation Range

ODU	Cooling (°F / °C)	-13 ~ 122 / -25 ~ 50
	Heating (°F / °C)	-13 ~ 75 / -25 ~ 24
IDU	Cooling (°F / °C)	60 ~ 90 / 16 ~ 32
	Heating (°F / °C)	32 ~ 86 / 0 ~ 30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		7.71
IDU	Fan motor model	ZKFN-400-8-1
	Fan motor qty (ea)	1
	Fan motor input (W)	427.0
	Fan speed, H / M / L (RPM)	790 / 730 / 670
	Air flow, Turbo / H / M / L / Si (CFM)	1236.1 / 1059.5 / 941.8 / 765.2 / N/A
	Noise level, Turbo / H / M / L / Si (dB(A))	N/A / 45 / 39.5 / 33 / 30
	Net weight (lbs / kg)	106.26 / 48.2
	Gross weight (lbs / kg)	120.37 / 54.6
ODU	ESP range (Pa)	0-200
	Drain pipe outside diameter in. (mm)	OD 0.98 (25)
	Fan motor model	ZKFN-120-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	170
	Fan motor speed, H / / L (RPM)	950 / 800 / 500
	Air flow, Max (CFM)	2413.3
	Noise level, Max (dB(A))	63.0
Accessories	BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
	BMS-DRY-XXD (8733969263)	24V Interface

Indoor Unit Dimensions and Connections

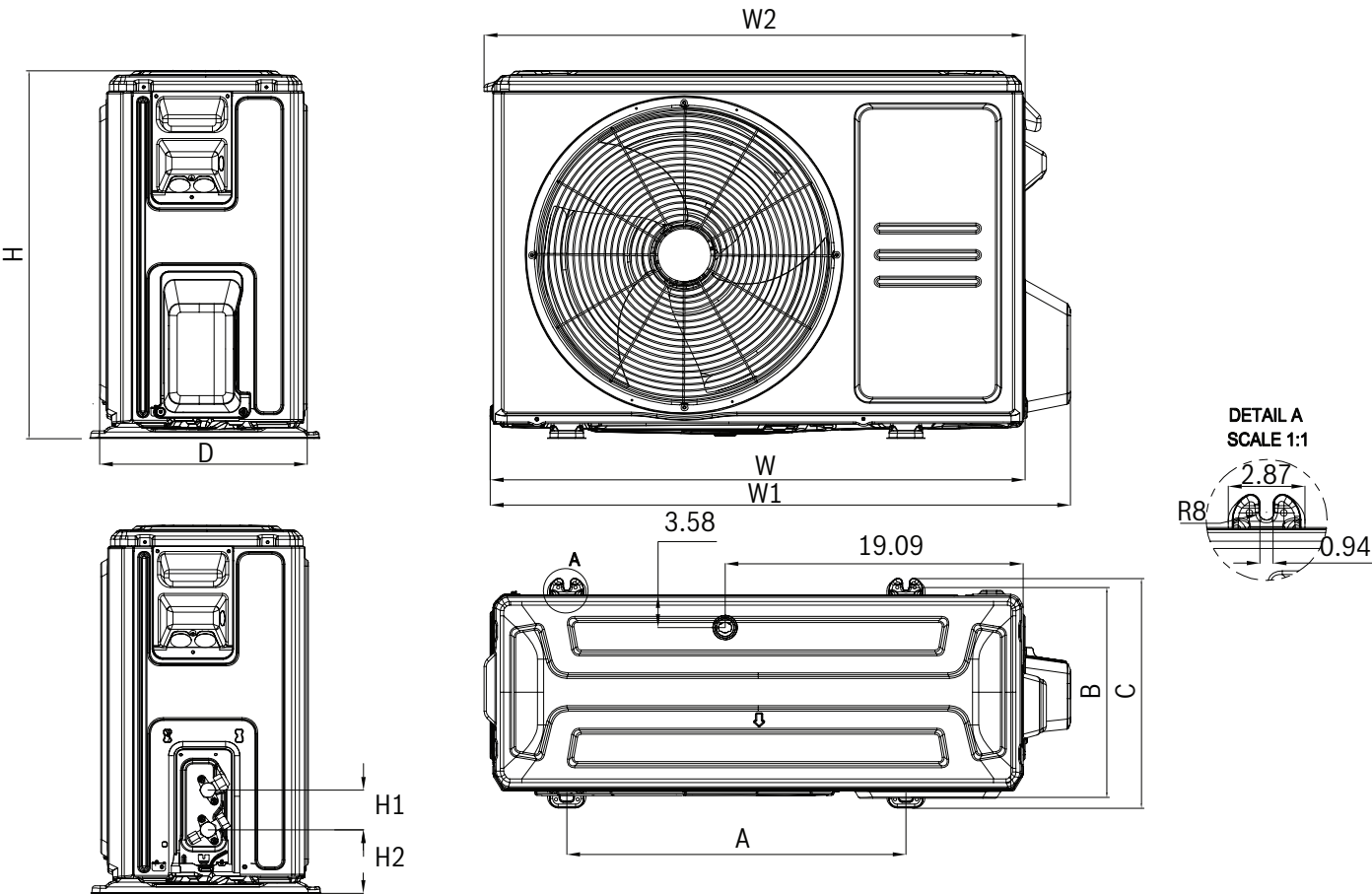


Model	Outline Dimension (in)				Air Outlet Opening Size (in)		Air Return Opening Size (in)		Size of Mounted Lug (in)		Fresh Air Intake Opening Size (in)	
	A	B	C	D	E	F	G	H	I	J	K	L
BMS500-AAU036-1AHDXD	47.2	11.8	29.5	31.3	40.4	9.2	43.0	10.5	48.8	25.2	4.9	6.3

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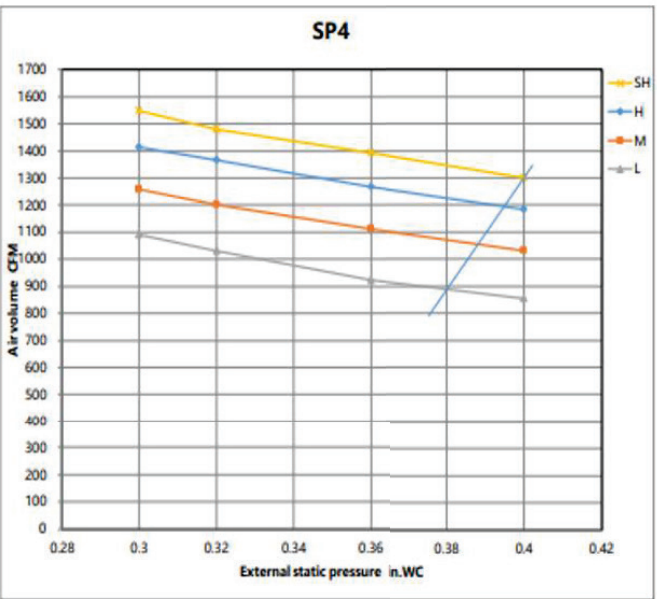
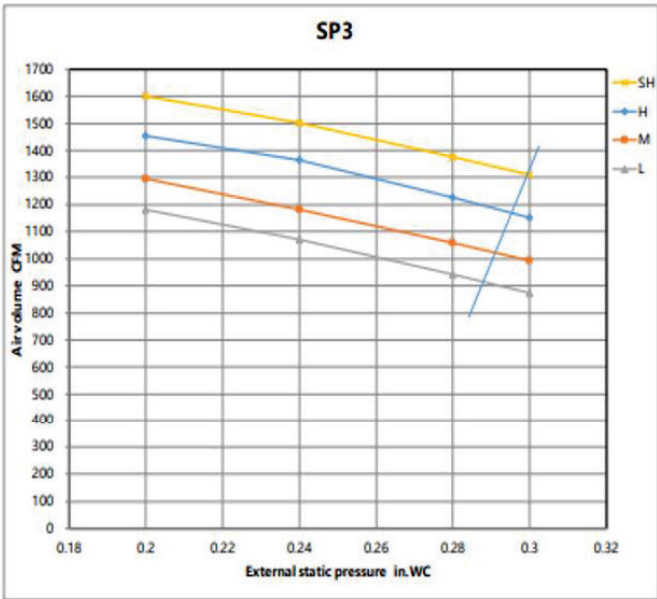
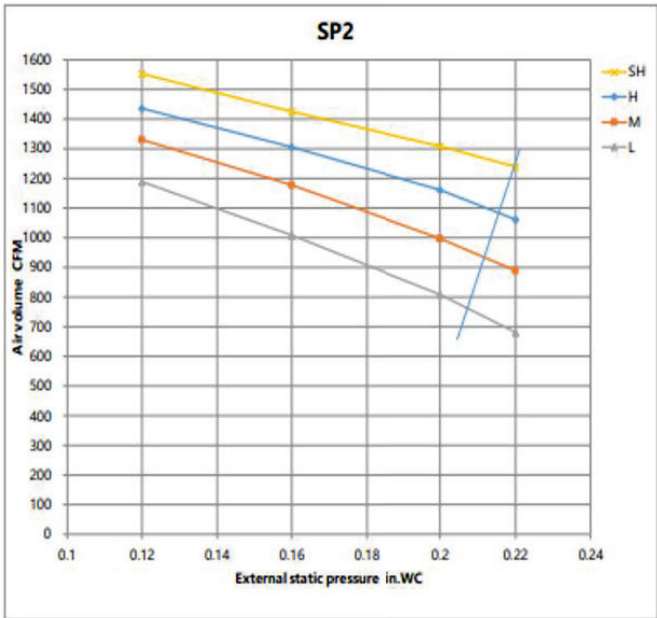
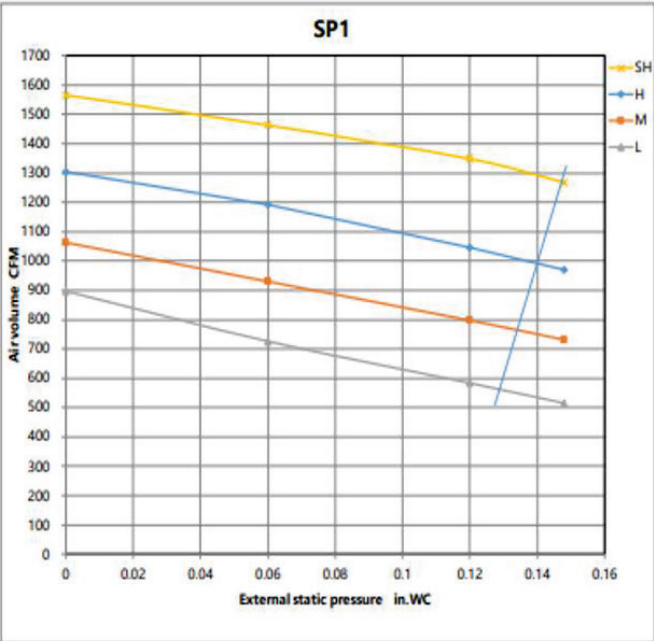


Outdoor Unit Dimensions and Connections

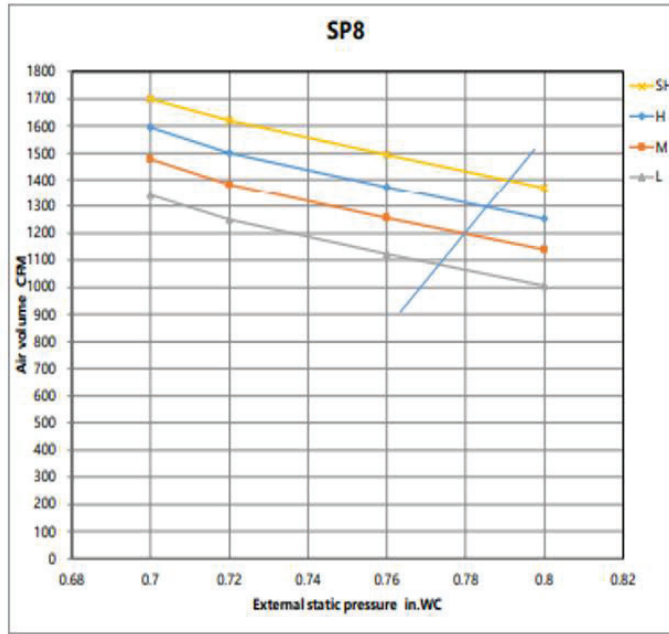
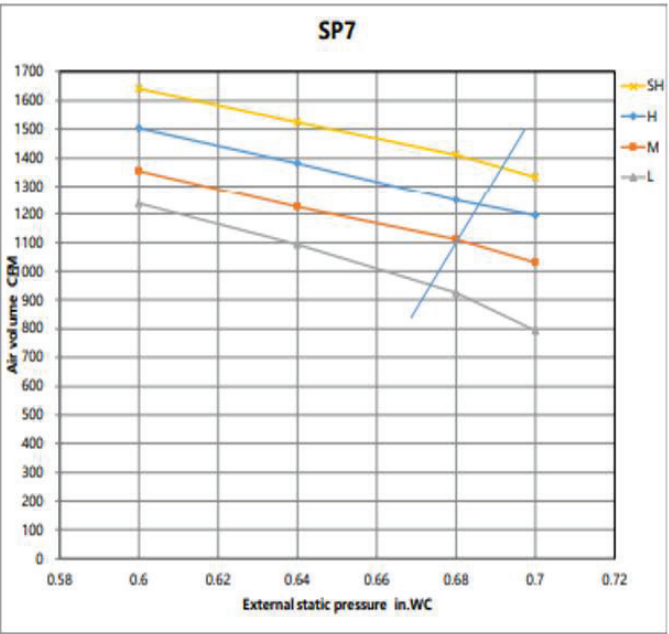
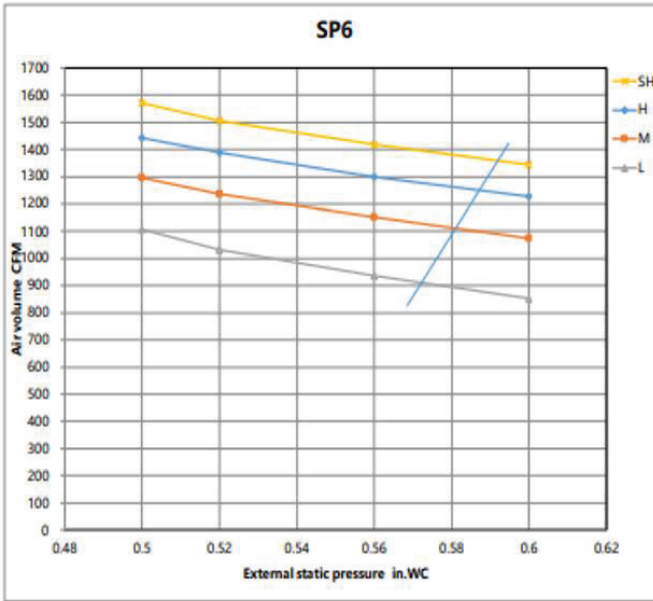
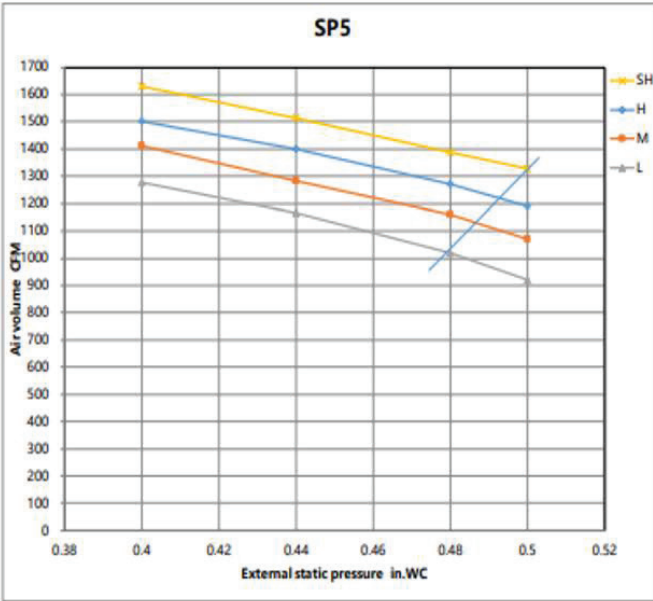


Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	H1 (in)	H2 (in)	A (in)	B (in)	C (in)
BMS500-AAS036-1CSXLD	37.24	16.14	31.89	40.55	37.24	2.36	4.25	26.5	16.14	17.9

Air Volume

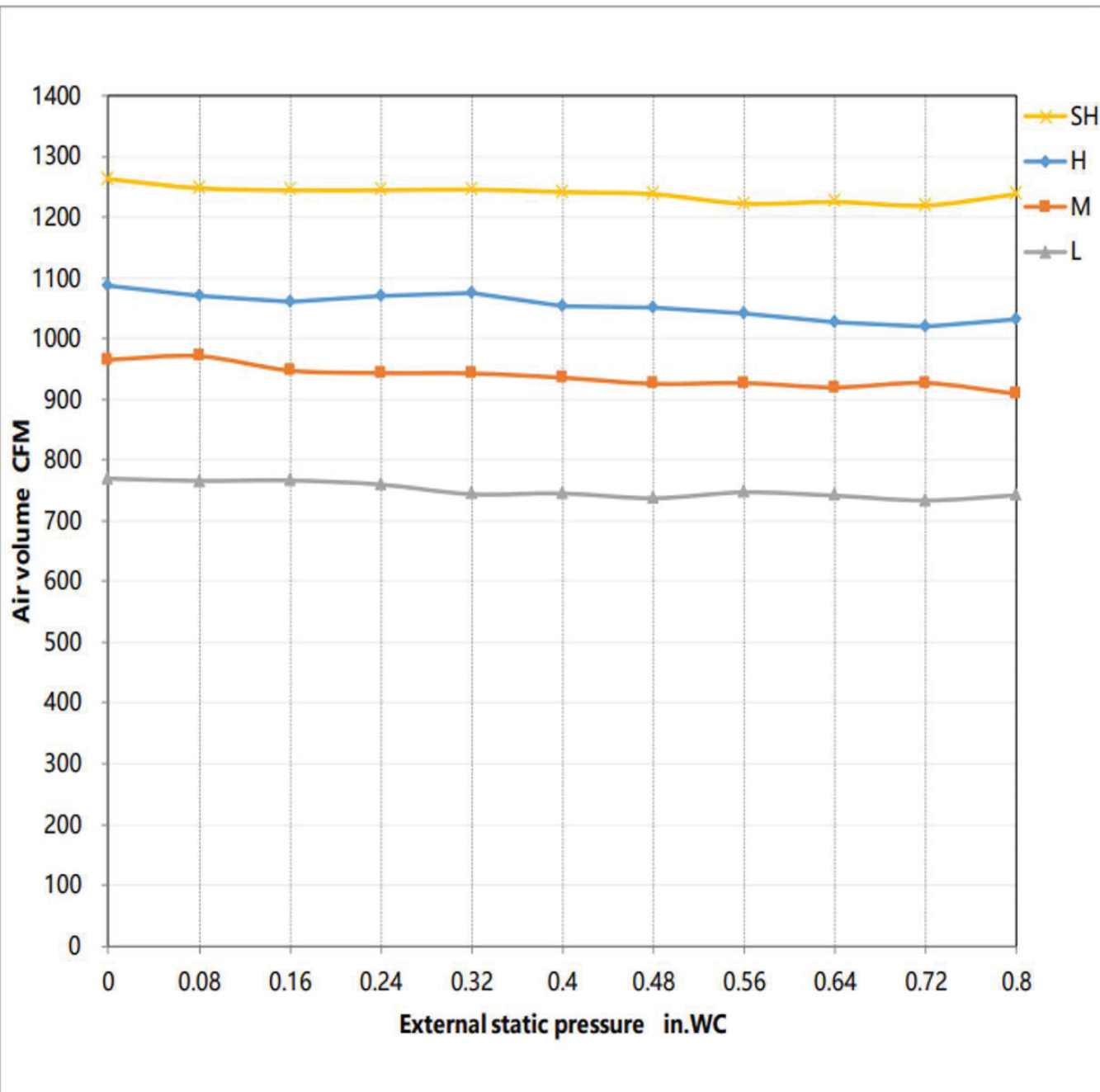


Air Volume



Air Volume

Constant air volume





Cooling Performance - Indoor Dry Bulb Temperature at 1235 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-13°F (-25.0°C)	TC	21261	21252	21555	21644	22240	22183	22165	22151	22149	22130	22846	22818	22796	22793	22771	24099	24092	24042	24033	23983
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	1.23	1.23	1.22	1.22	1.22	1.22	1.22	1.22	1.22	1.22	1.22	1.22	1.22	1.22	1.22	1.23	1.23	1.23	1.23	1.23
-4°F (-20.0°C)	TC	22897	22886	23213	23309	23951	23890	23870	23855	23853	23833	24604	24573	24550	24546	24523	25952	25945	25892	25882	25827
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	1.27	1.27	1.27	1.27	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.27	1.27	1.27	1.27	1.27
0°F (-17.8°C)	TC	23657	23643	24081	24182	24846	24671	24644	24625	24622	24870	25396	25360	25329	25325	25313	26762	26756	26754	26757	26775
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	1.28	1.28	1.28	1.28	1.28	1.28	1.28	1.28	1.28	1.28	1.28	1.28	1.28	1.28	1.28	1.30	1.30	1.30	1.30	1.30
5°F (-15.0°C)	TC	24335	24419	24977	25092	25865	25476	25483	25484	25511	25891	26291	26298	26303	26304	26339	27833	27825	27839	27842	27854
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	1.29	1.29	1.29	1.29	1.30	1.30	1.30	1.30	1.30	1.30	1.31	1.31	1.31	1.31	1.31	1.33	1.33	1.33	1.33	1.34
14°F (-10.0°C)	TC	22651	22640	23102	23211	23930	23776	23755	23739	23736	23958	24591	24560	24535	24531	24506	26148	26140	26089	26080	26025
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	1.29	1.29	1.28	1.28	1.28	1.28	1.28	1.28	1.28	1.28	1.28	1.28	1.28	1.28	1.28	1.29	1.29	1.29	1.29	1.29
17°F (-8.3°C)	TC	22040	22112	22623	22733	23472	23137	23147	23151	23170	23497	23927	23936	23945	23946	23976	25447	25440	25436	25441	25462
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.28	1.28	1.28	1.28	1.28
23°F (-5.0°C)	TC	22702	22800	23390	23505	24280	23852	23859	23862	23893	24306	24674	24683	24689	24690	24730	26236	26228	26247	26250	26266
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	1.21	1.21	1.21	1.22	1.22	1.22	1.22	1.22	1.22	1.22	1.23	1.23	1.23	1.23	1.23	1.25	1.25	1.25	1.25	1.25
35°F (1.7°C)	TC	24249	24250	24393	24461	25250	25383	25382	25384	25384	25452	26177	26176	26177	26178	26180	27651	27645	27643	27642	27645
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	1.03	1.03	1.03	1.04	1.08	1.09	1.09	1.09	1.09	1.09	1.13	1.13	1.13	1.14	1.14	1.23	1.23	1.23	1.23	1.23
41°F (5.0°C)	TC	24353	24354	24548	24718	25449	25486	25486	25487	25487	25576	26281	26279	26281	26281	26283	27756	27749	27747	27746	27749
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	1.08	1.08	1.09	1.10	1.15	1.15	1.15	1.15	1.15	1.15	1.20	1.20	1.20	1.20	1.20	1.31	1.31	1.31	1.31	1.31
47°F (8.3°C)	TC	24457	24473	24801	24912	25649	25591	25590	25591	25590	25709	26386	26383	26385	26385	26385	27863	27856	27853	27852	27855
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	1.14	1.14	1.16	1.17	1.22	1.21	1.21	1.21	1.21	1.22	1.27	1.27	1.27	1.27	1.27	1.38	1.38	1.38	1.38	1.38
50°F (10.0°C)	TC	24057	24720	25554	25702	26698	25415	25429	25605	25686	26724	26407	26441	26455	26457	26711	28227	28302	28374	28383	28419
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	1.30	1.30	1.30	1.30	1.30	1.30	1.30	1.30	1.30	1.30	1.30	1.30	1.30	1.30	1.30	1.30	1.29	1.29	1.29	1.29
59°F (15.0°C)	TC	24171	25019	25907	26065	27126	25575	25609	25900	26093	27152	26630	26656	26666	26705	27175	28595	28652	28709	28716	28740
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	1.37	1.37	1.38	1.38	1.38	1.38	1.38	1.38	1.38	1.38	1.38	1.38	1.38	1.38	1.38	1.39	1.39	1.39	1.39	1.39
68°F (20.0°C)	TC	24273	25309	26248	26415	27535	25697	25755	26276	26443	27563	26808	26828	26876	26900	27585	28885	28939	28984	28989	29004
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	1.44	1.45	1.46	1.46	1.47	1.46	1.46	1.46	1.46	1.47	1.47	1.47	1.47	1.47	1.47	1.49	1.49	1.48	1.48	1.48
77°F (25.0°C)	TC	24354	25526	26525	26702	27895	25818	25900	26554	26732	27923	26987	27018	27087	27117	27947	29147	29236	29303	29311	29338
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	1.53	1.54	1.55	1.55	1.56	1.54	1.54	1.55	1.55	1.56	1.55	1.55	1.55	1.55	1.56	1.57	1.57	1.57	1.57	1.56
82°F (27.8°C)	TC	24510	25762	26801	26982	28218	25973	26074	26824	27013	28248	27182	27216	27301	27334	28272	29389	29500	29576	29584	29617
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	1.71	1.72	1.72	1.72	1.73	1.72	1.72	1.72	1.72	1.73	1.73	1.73	1.73	1.73	1.73	1.75	1.74	1.74	1.74	1.74
86°F (30.0°C)	TC	24862	26103	27171	27360	28635	26219	26340	27201	27392	28664	27462	27493	27594	27634	28689	29730	29845	29916	29923	29951
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	2.00	2.01	2.02	2.03	2.04	2.02	2.02	2.02	2.03	2.04	2.03	2.03	2.03	2.03	2.04	2.06	2.05	2.05	2.05	2.05
95°F (35.0°C)	TC	25471	25608	26436	26592	27643	27070	27086	27094	27126	27678	28231	28237	28255	28257	28304	30421	30409	30415	30424	30465
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.58	0.68
	PI	2.57	2.58	2.62	2.62	2.68	2.65	2.65	2.65	2.65	2.68	2.71	2.71	2.71	2.71	2.71	2.82	2.82	2.82	2.82	2.82
104°F (40.0°C)	TC	21655	21853	22676	22812	23722	23019	23032	23073	23098	23752	24002	24011	24024	24025	24093					

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BOSCH

Heating Performance - Indoor Dry Bulb Temperature at 1235 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-13°F(-25.0°C)	17874	17516	17428	17359	17203	3.18	3.20	3.21	3.21	3.23
-4°F(-20.0°C)	21968	21528	21421	21335	21143	3.33	3.35	3.36	3.37	3.38
0°F(-17.8°C)	24269	23784	23665	23570	23358	3.40	3.43	3.44	3.44	3.46
5°F(-15.0°C)	27146	26603	26470	26364	26127	3.50	3.53	3.53	3.54	3.55
14°F(-10.0°C)	30597	29985	29835	29716	29449	3.70	3.73	3.74	3.74	3.76
17°F(-8.3°C)	31748	31113	30957	30833	30556	3.77	3.80	3.80	3.81	3.83
22°F(-5.6°C)	33402	32734	32570	32440	32148	3.72	3.75	3.75	3.76	3.78
27°F(-2.8°C)	35056	34355	34183	34047	33740	3.67	3.70	3.70	3.71	3.73
32°F(0.0°C)	36711	35976	35796	35653	35332	3.62	3.65	3.65	3.66	3.68
35°F(1.7°C)	37703	36949	36764	36617	36288	3.59	3.62	3.63	3.63	3.65
37°F(2.8°C)	38826	38050	37860	37708	37369	3.54	3.57	3.58	3.59	3.60
42°F(5.6°C)	41634	40802	40598	40435	40071	3.44	3.46	3.47	3.48	3.49
44.6°F(7.0°C)	42758	41902	41693	41526	41152	3.39	3.42	3.43	3.43	3.45
47°F(8.3°C)	44442	43554	43336	43162	42774	3.33	3.35	3.36	3.37	3.38
52°F(11.1°C)	45900	44982	44757	44578	44177	3.16	3.18	3.19	3.19	3.21
57°F(13.9°C)	47358	46411	46179	45994	45580	2.99	3.01	3.02	3.02	3.04
59°F(15.0°C)	47942	46983	46748	46561	46142	2.92	2.94	2.95	2.95	2.97
62°F(16.7°C)	48816	47840	47601	47410	46984	2.82	2.84	2.85	2.85	2.86
64.4°F(18.0°C)	49516	48526	48283	48090	47657	2.74	2.76	2.76	2.77	2.78
69.8°F(21.0°C)	46321	45395	45168	44987	44582	2.55	2.57	2.58	2.58	2.59
75.2°F(24.0°C)	32745	32090	31929	31802	31516	2.37	2.39	2.39	2.40	2.41

Bolded values were tested. All other values are simulated and for reference only.